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Line-of-Sight Density and Hubble Residuals in Pantheon: The CET Non-Acceleration Interpretation of Redshift

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ABSTRACT

We present an exploratory test of whether the integrated line-of-sight (LOS) density correlates with SN Ia Hubble residuals in the Pantheon+SH0ES sample cross-matched with 2MRS. Two distance prescriptions are compared: (i) the standard ΛCDM luminosity distance and (ii) a linear distance motivated by the Cosmic Elastic Theory (CET). A weak but significant correlation is found under ΛCDM (Spearman $\rho \approx -0.17$, $p \sim 2 \times 10^{-4}$), which becomes stronger under CET ($\rho \approx -0.23$, $p \sim 5 \times 10^{-7}$). The signal persists after redshift detrending and in partial correlation controlling for z. These results suggest that residuals in the Hubble diagram carry an environmental dependence, consistent with interpretations where light propagation includes a dissipative component. All data, figures and analysis scripts are released for reproducibility.

Keywords: supernovae: general — cosmology: observations — large-scale structure of universe

1. OBJECTIVE

The goal is to test whether the integrated line-of-sight (LOS) density up to the redshift of the SN correlates with Hubble residuals, and whether this correlation depends on the distance anchor. We compare standard ΛCDM distances with a CET-inspired linear distance.

2. METHODOLOGY

2.1. *Sample*

We start from Pantheon+SH0ES (1701 SNe Ia). After cross-matching with 2MRS, 492 SNe have at least one identified neighbor. Requiring a valid LOS density score leaves 205 SNe for the binned LOS analysis. For the redshift-corrected analysis, after quality cuts the working sample is 466 SNe.

2.2. *LOS score*

The LOS density score is defined as the integrated density along the path to $z_{\rm SN}$:

$$S_{\rm LOS} = \int_0^{z_{\rm SN}} w(z)\,\rho(z)\,dz, \tag{1}$$

with $\rho(z)$ from 2MRS and $w(z)$ a geometric weight. In practice, $S_{\rm LOS}$ is computed by summing density in redshift bins with an angular mask around each SN.

2.3. *Distances and residuals*

For each SN we compute theoretical distance modulus and residual $\Delta\mu = \mu_{\rm obs} - \mu_{\rm th}$ under two anchors:

ΛCDM.—

$$D_L(z) = \frac{c(1+z)}{H_0}\int_0^z \frac{dz'}{\sqrt{\Omega_m(1+z')^3+\Omega_\Lambda}}, \quad \mu = 5\log_{10}(D_L/10\,{\rm pc} \tag{2}$$

CET (linear).—

$$D_L^{\rm CET}(z) = \frac{c}{H_0}\,z, \qquad \mu_{\rm CET} = 5\log_{10}(D_L^{\rm CET}/10\,{\rm pc}). \tag{3}$$

2.4. *Statistics*

We compute Spearman correlation between $\Delta\mu$ and $S_{\rm LOS}$, with uncertainties via bootstrap and jackknife. We also perform binned analysis, redshift detrending, and partial correlations (LOS vs. $\Delta\mu$ controlling for z).

3. RESULTS

3.1. *ΛCDM*

We find $\rho \approx -0.17$ (N=466, $p \approx 2.3 \times 10^{-4}$). In binned analysis $\langle\Delta\mu\rangle$ decreases from low to high LOS. After redshift correction, the signal weakens ($\rho \approx -0.12$, $p \approx 0.011$) but remains marginally significant. Partial correlation yields $\rho \approx -0.19$ ($p \sim 2.6 \times 10^{-5}$).

3.2. *CET*

Under CET distances the correlation strengthens: $\rho \approx -0.23$ (N=466, $p \approx 4.8 \times 10^{-7}$). The binned trend is clearer, and both detrended ($\rho \approx -0.13$, $p \approx 6.8 \times 10^{-3}$)

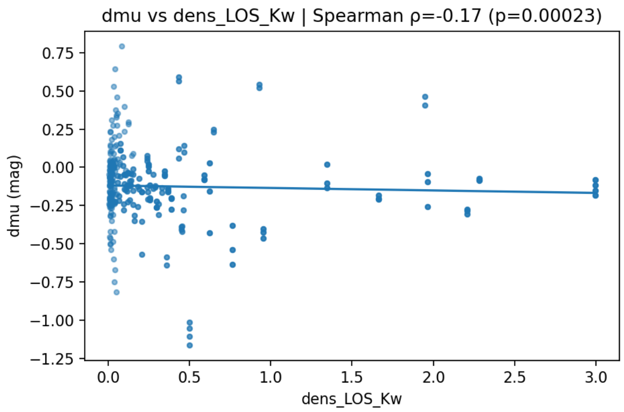

Figure 1. Hubble residuals $\Delta\mu$ (LCDM) vs. LOS score.

Table 1. Binned means of $\Delta\mu$ by LOS terciles (LCDM).

LOS bin	N	$\langle\Delta\mu\rangle$	SEM	$\langle z\rangle$
low	155	-0.083	0.018	0.026
mid	161	-0.120	0.013	0.021
high	150	-0.178	0.027	0.011

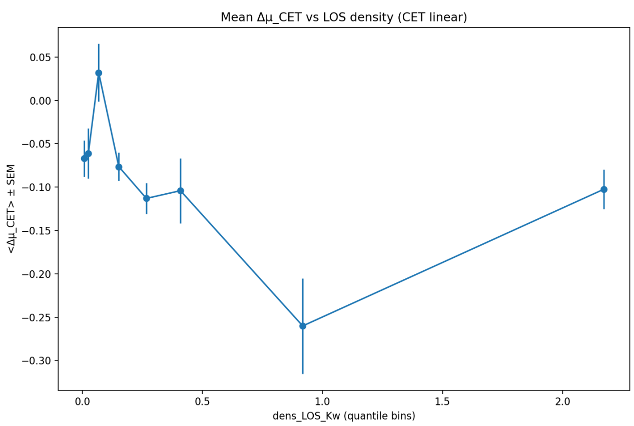

Figure 2. Hubble residuals $\Delta\mu$ (CET) vs. LOS score.

and partial correlations ($\rho \approx -0.20$, $p \approx 1.1 \times 10^{-5}$) remain highly significant.

3.3. *Distance correction in CET logic*

In CET, part of the observed redshift reflects elastic dissipation. Recasting distance as linear at low-z removes part of the geometric mismatch and emphasizes the physical LOS dependence. The improvement in significance (ΛCDM $\to$ CET) is consistent with this picture.

3.4. *Density–redshift relation*

With explicit normalization by redshift, the LOS–$\Delta\mu$ correlation persists (and is more stable under CET). This indicates the signal is not a redshift-distribution artifact.

4. CONCLUSION

Cross-matching Pantheon SNe with 2MRS reveals a reproducible correlation between LOS density and Hubble residuals. The correlation is present under standard ΛCDM distances and becomes stronger when a CET-motivated linear distance is adopted. Importantly, the signal remains significant after controlling for redshift. These findings highlight an environmental imprint on SN Ia residuals that deserves further investigation with larger and independent samples. While no definitive interpretation is claimed here, the results are consistent with the possibility that light propagation has a dissipative component. All products (data, figures, and scripts) are made openly available to facilitate independent verification and future extensions.

AUTHOR NOTE

This version is released as a public preprint under OSF (https://osf.io/vpq76) for open access and feedback. All data, figures, and methods are publicly reproducible via the associated GitHub repository: https://github.com/LeonardoSeriacopi

APPENDIX

A. LOS CALCULATION DETAILS

We specify angular radius, Δz criterion and weight $w(z)$ as: [fill in]. Robustness checks: jackknife by sky region, variation of parameters, randomizations.

A.1. *Additional figures*

Table 2. Quantile-binned means of $\Delta\mu_{\mathrm{CET}}$ vs LOS (CET).

Bin left	Bin right	N	$\langle\Delta\mu_{\mathrm{CET}}\rangle$	SEM
0.003	0.013	59	-0.067	0.021
0.013	0.038	58	-0.061	0.029
0.038	0.099	60	$+0.032$	0.033
0.099	0.205	58	-0.076	0.016
0.205	0.329	59	-0.113	0.018
0.329	0.489	55	-0.104	0.037
0.489	1.347	61	-0.260	0.055
1.347	2.996	56	-0.102	0.023

Table 3. LOS–residual correlations (Spearman).

Anchor	N	ρ (raw)	p	ρ (partial $\|z$)
ΛCDM	466	-0.17	2.3×10^{-4}	-0.19 (2.6×10^{-5})
CET	466	-0.23	4.8×10^{-7}	-0.20 (1.1×10^{-5})

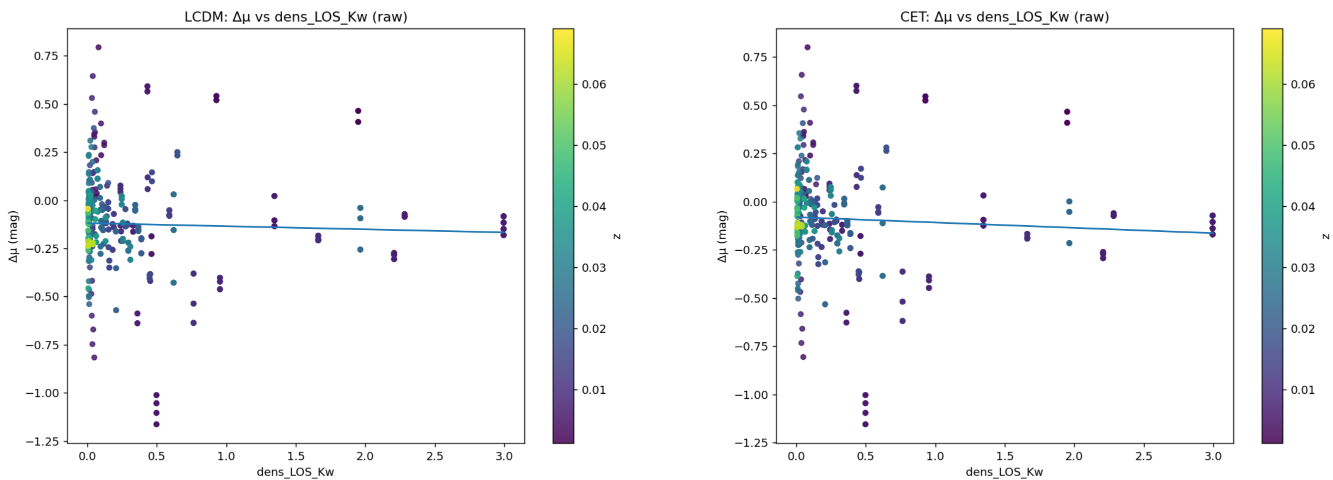

Figure 3. Raw $\Delta\mu$ vs. S_{LOS} for ΛCDM (left) and CET (right), with best-fit linear trends overplotted.

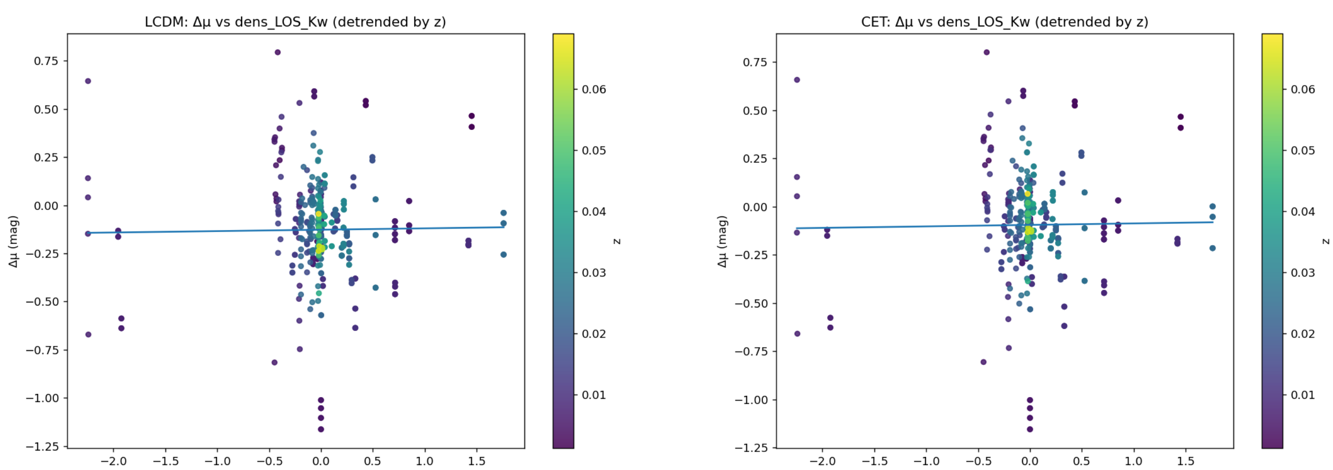

Figure 4. z-detrended $\Delta\mu$ vs. S_{LOS} for ΛCDM (left) and CET (right). Same data as Fig. **??**/**??**, shown here for completeness.

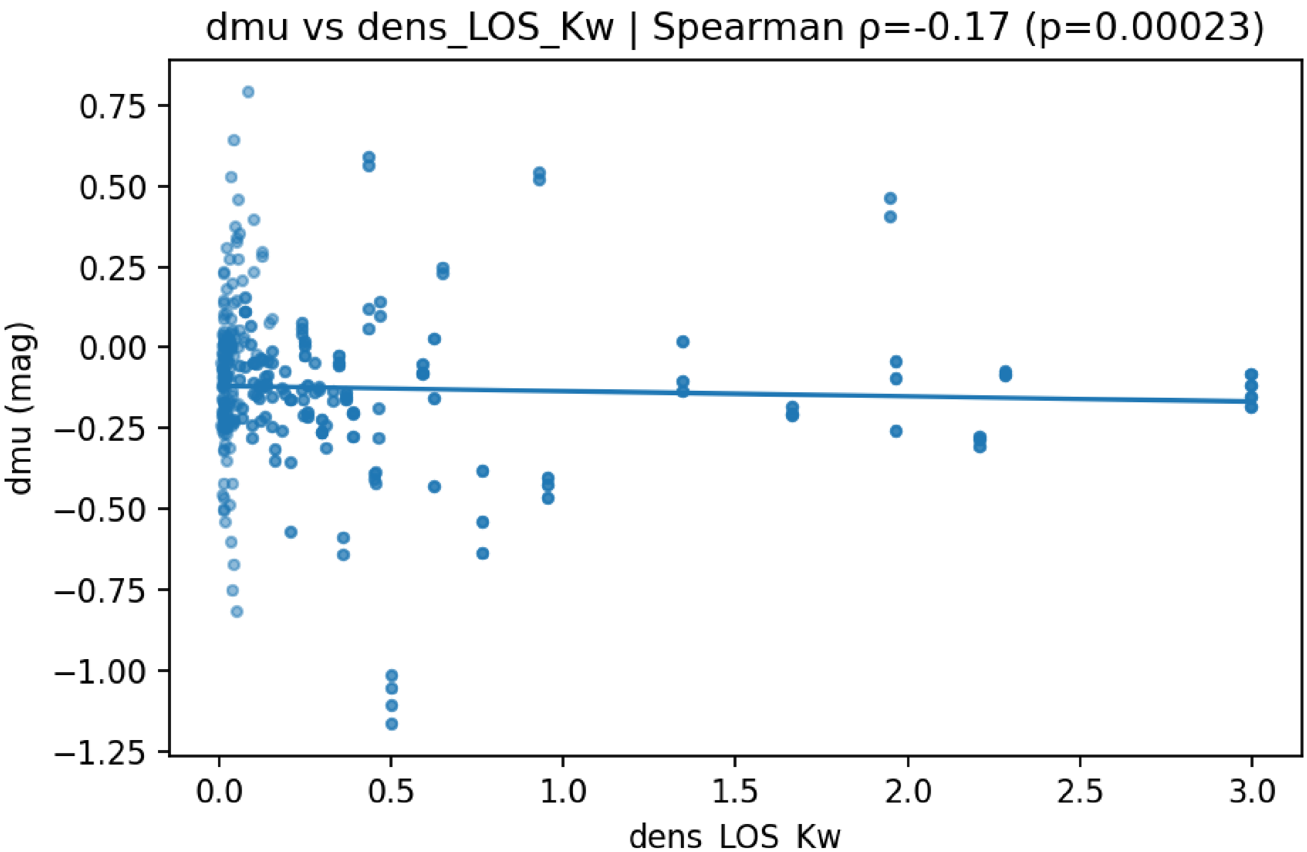

Figure 5. ΛCDM: $\Delta\mu$ vs. LOS density scatter with linear trend (alternate rendering).

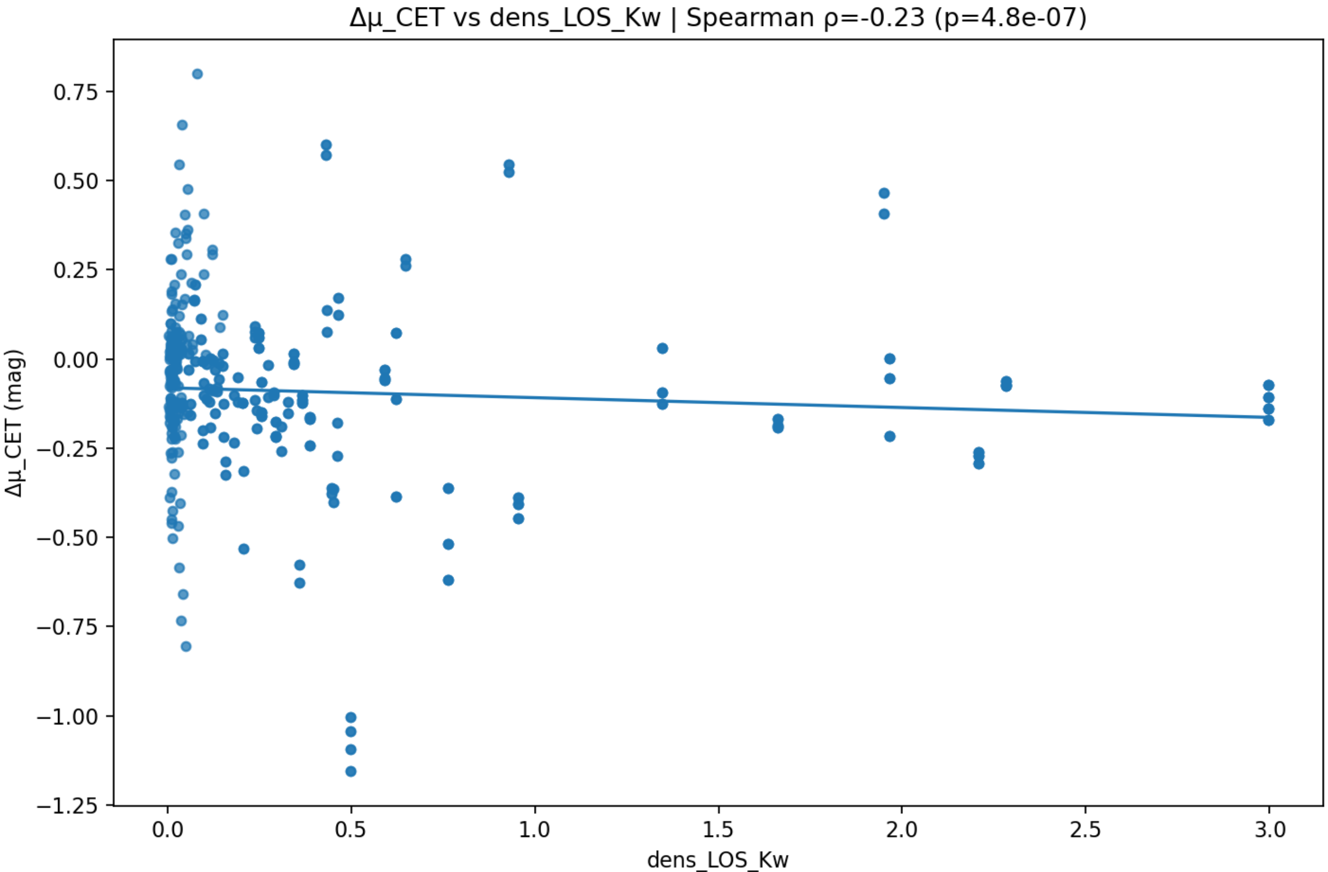

Figure 6. CET: $\Delta\mu_{\rm CET}$ vs. LOS density scatter with linear trend (alternate rendering).

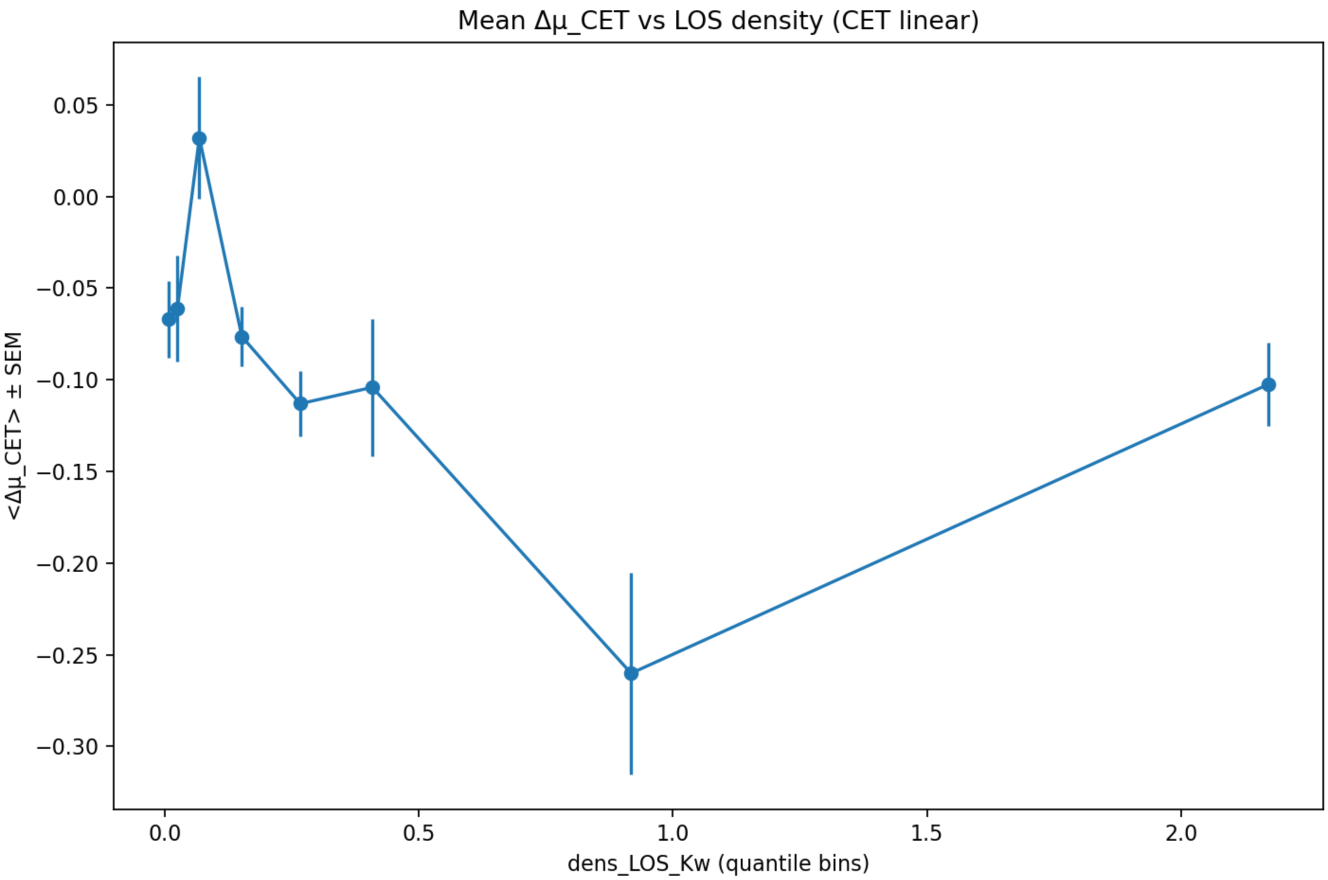

Figure 7. CET: quantile-binned means of $\langle \Delta\mu_{\mathrm{CET}} \rangle$ vs. LOS density (error bars = SEM).